

Akilia Mathie

Data Scientist / Data Engineer

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SUMMARY

Data Scientist with experience developing scalable data systems, machine-learning workflows, and analytical solutions across healthcare and technology-driven environments. Skilled in Python, SQL, and cloud platforms (Azure, AWS, Snowflake) for ETL pipeline development, data modeling, and large-scale data processing. Strong in statistical analysis, feature engineering, data visualization, and workflow automation. Known for improving data quality, resolving cross-system discrepancies, and communicating insights effectively to both technical and non-technical stakeholders.

TECHNICAL SKILLS

Programming & Analytics

Python (Pandas, NumPy, Scikit-learn, PyCaret), R, SQL (PostgreSQL, MySQL, Access, Hive/Impala), Git, Bash, Docker

Data Engineering

ETL/ELT Pipelines, Data Modeling, Workflow Automation (Snakemake, Nextflow), Data Ingestion, Data Transformation, Scalable Processing, Distributed Systems

Cloud & Big Data

Azure, AWS, Snowflake, BigQuery, Hive/Impala, Parallel Query Processing, Cloud Storage/Data Lakes

Visualization & Reporting

Tableau, Power BI, R Shiny

Data Quality & Governance

QC/QA, Data Cleaning, Anomaly Detection, Metadata Management, Reproducibility, Documentation

CERTIFICATIONS

<i>The Complete SQL Bootcamp – Udemy (Jose Portilla)</i>	<i>Python Machine Learning Bootcamp – Accelebrate</i>
<i>Python Programming Bootcamp – Accelebrate</i>	<i>R Programming Bootcamp – Accelebrate</i>

PROFESSIONAL EXPERIENCE

Data Scientist Centers for Disease Control and Prevention (CDC) – *Jan 2023 – Present*

- Designed and maintained scalable ETL pipelines using Python, SQL, and R to ingest, transform, and validate data across Azure, Snowflake, PostgreSQL, MySQL, Access, and Hive/Impala.
- Developed automated data-quality and validation workflows in Python to detect anomalies, missingness, duplicates, and cross-system discrepancies.
- Built predictive and statistical models using Python (Scikit-learn, PyCaret) to support forecasting, clustering, and analytical decision-making.
- Applied feature engineering and data preprocessing techniques to improve model performance and analytical accuracy.
- Performed advanced SQL querying and feature engineering on large-scale datasets to prepare model-ready inputs and support downstream analytics.
- Integrated and standardized data from diverse sources and formats, ensuring consistency and reliability across systems.
- Created interactive dashboards and reports in Tableau and Power BI to visualize trends, model outputs, and data-quality metrics for technical and non-technical stakeholders.
- Collaborated cross-functionally with clinicians, epidemiologists, engineers, and leadership to translate analytical findings into actionable insights and workflow improvements.
- Implemented reproducible analytical workflows using version control, documentation, and structured metadata to ensure transparency, traceability, and reliability.

Quality Control Specialist, R&D (Contract) Immucor – *Jul 2022 – Jan 2023*

- Performed statistical QC on assay and manufacturing data using SQL and Python.
- Built automated dashboards for trend analysis, outlier detection, and technician performance monitoring.
- Reduced manual reporting effort by 40% through workflow automation and standardized reporting pipelines.

Early Experience: Molecular Technologist (Tempus) *Jan 2022 – Jun 2022* | Lab Technician (Grady Hospital) *Jan 2021 – Dec 2021* | Pharmacy Technician (Walgreens) *Sep 2017 – Jan 2021*

EDUCATION

Master of Science in Biotechnology – Morehouse School of Medicine, Atlanta, GA

Bachelor of Science in Molecular Biology – Kennesaw State University, Kennesaw, GA